| | |
|---|---|
| **Status** | Finished |
| **Started** | Thursday, 10 September 2026, 8:48 AM |
| **Completed** | Thursday, 10 September 2026, 9:21 AM |
| **Duration** | 33 mins 46 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid.

An input string is valid if:

Open brackets must be closed by the same type of brackets.

Open brackets must be closed in the correct order.

Constraints:

1 <= s.length <= 10^4

s consists of parentheses only '()[]{}'.

**For example:**

| Test | Result |
|---|---|
| `print(ValidParenthesis("()"))` | `true` |
| `print(ValidParenthesis("()[]{}"))` | `true` |
| `print(ValidParenthesis("(]"))` | `false` |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def ValidParenthesis(s):
    stack=[]
    mapping={")":"(","}":"{","]":"["}
    for char in s:
        if char in mapping:
            top_element=stack.pop() if stack else "#"
            if mapping[char]!=top_element:
                return "false"
        else:
            stack.append(char)
    return "true" if not stack else "false"

```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | `print(ValidParenthesis("()"))` | `true` | `true` | ✓ |
| ✓ | `print(ValidParenthesis("()[]{}"))` | `true` | `true` | ✓ |
| ✓ | `print(ValidParenthesis("(]"))` | `false` | `false` | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Given a **non-empty** string s and an abbreviation abbr, return whether the string matches with the given abbreviation.

A string such as "word" contains only the following valid abbreviations:

["word", "1ord", "w1rd", "wo1d", "wor1", "2rd", "w2d", "wo2", "1o1d", "1or1", "w1r1", "1o2", "2r1", "3d", "w3", "4"]

Notice that only the above abbreviations are valid abbreviations of the string "word". Any other string is not a valid abbreviation of "word".

**Note:**

Assume s contains only lowercase letters and abbr contains only lowercase letters and digits.

**Example 1:**

**Input**

internationalization
i12iz4n

**Output**

true

**Explanation**

Given **s** = "internationalization", **abbr** = "i12iz4n":

Return true.

**Example 2:**

**Input**

apple
a2e

**Output**

false

**Explanation**

Given **s** = "apple", **abbr** = "a2e":

Return false.

**Answer:** (penalty regime: 0 %)

```
def valid_word_abbreviation(s, abbr):
    i = 0
    j = 0

    while i < len(s) and j < len(abbr):

        if abbr[j].isalpha():
            if s[i] != abbr[j]:
                return False

            i += 1
            j += 1

        else:

            if abbr[j] == '0':
                return False

            num = 0

            while j < len(abbr) and abbr[j].isdigit():
                num = num * 10 + int(abbr[j])
                j += 1

            i += num

            if i > len(s):
                return False

    return i == len(s) and j == len(abbr)



s = input().strip()
abbr = input().strip()


if valid_word_abbreviation(s, abbr):
    print("true")
else:
    print("false")

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | internationalization<br>i12iz4n | true | true | ✓ |
| ✓ | apple<br>a2e | false | false | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given a string S which is of the format USERNAME@DOMAIN.EXTENSION, the program must print the EXTENSION, DOMAIN, USERNAME in the reverse order.

**Input Format:**

The first line contains S.

**Output Format:**

The first line contains EXTENSION.
The second line contains DOMAIN.
The third line contains USERNAME.

**Boundary Condition:**

1 <= Length of S <= 100

**Example Input/Output 1:**

Input:

abcd@gmail.com

Output:

com
gmail
abcd

**For example:**

| Input | Result |
|---|---|
| arvijayakumar@rajalakshmi.edu.in | edu.in<br>rajalakshmi<br>arvijayakumar |

**Answer:** (penalty regime: 0 %)

```
S = input().strip()


username, domain = S.split('@')


domain_name, extension = domain.split('.', 1)

print(extension)
print(domain_name)
print(username)

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | abcd@gmail.com | com<br>gmail<br>abcd | com<br>gmail<br>abcd | ✓ |
| ✓ | arvijayakumar@rajalakshmi.edu.in | edu.in<br>rajalakshmi<br>arvijayakumar | edu.in<br>rajalakshmi<br>arvijayakumar | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Assume that the given string has enough memory.

Don't use any extra space(IN-PLACE)

**Sample Input 1**

a2b4c6

**Sample Output 1**

aabbbbccccccc

**Answer:** (penalty regime: 0 %)

```
s = input().strip()

result = []
i = 0

while i < len(s):
    ch = s[i]
    i += 1

    count = 0

    while i < len(s) and s[i].isdigit():
        count = count * 10 + int(s[i])
        i += 1

    result.extend([ch] * count)

print(''.join(result))

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | a2b4c6 | aabbbbccccccc | aabbbbccccccc | ✓ |
| ✓ | a12b3d4 | aaaaaaaaaaaabbbdddd | aaaaaaaaaaaabbbdddd | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Write a Python program to get one string and reverses a string. The input string is given as an array of characters `char[]`.

You may assume all the characters consist of printable ascii characters.

**Example 1:**

```
Input:
hello
Output:
olleh
```

**Example 2:**

```
Input:
Hannah
Output:
hannaH
```

**Answer:** (penalty regime: 0 %)

```
s = list(input().strip())

left = 0
right = len(s) - 1

while left < right:
    s[left], s[right] = s[right], s[left]
    left += 1
    right -= 1

print(''.join(s))
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | hello | olleh | olleh | ✓ |
| ✓ | Hannah | hannaH | hannaH | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Consider the below words as key words and check the given input is key word or not.

keywords: {break, case, continue, default, defer, else, for, func, goto, if, map, range, return, struct, type, var}

Input format:

Take string as an input from stdin.

Output format:

Print the word is key word or not.

Example Input:

break

Output:

break is a keyword

Example Input:

IF

Output:

IF is not a keyword

**For example:**

| Input | Result |
|---|---|
| break | break is a keyword |
| IF | IF is not a keyword |

**Answer:** (penalty regime: 0 %)

```
word = input().strip()

keywords = {
    "break", "case", "continue", "default", "defer",
    "else", "for", "func", "goto", "if", "map",
    "range", "return", "struct", "type", "var"
}

if word in keywords:
    print(word + " is a keyword")
else:
    print(word + " is not a keyword")

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | break | break is a keyword | break is a keyword | ✓ |
| ✓ | IF | IF is not a keyword | IF is not a keyword | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Given a string, determine if it is a palindrome, considering only alphanumeric characters and ignoring cases.

**Note:** For the purpose of this problem, we define empty string as valid palindrome.

**Example 1:**

```
Input:
A man, a plan, a canal: Panama

Output:
1
```

**Example 2:**

```
Input:
race a car

Output:
0
```

**Constraints:**

- `s` consists only of printable ASCII characters.

**Answer:** (penalty regime: 0 %)

```
s = input()

# Keep only alphanumeric characters and convert to lowercase
clean = ""

for ch in s:
    if ch.isalnum():
        clean += ch.lower()

# Check palindrome
if clean == clean[::-1]:
    print(1)
else:
    print(0)
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | A man, a plan, a canal: Panama | 1 | 1 | ✓ |
| ✓ | race a car | 0 | 0 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

A pangram is a sentence where every letter of the English alphabet appears at least once.

Given a string sentence containing only lowercase English letters, return true if sentence is a pangram, or false otherwise.

Example 1:

Input:

thequickbrownfoxjumpsoverthelazydog

Output:

true

Explanation: sentence contains at least one of every letter of the English alphabet.

Example 2:

Input:

arvijayakumar

Output: false

Constraints:

1 <= sentence.length <= 1000

sentence consists of lowercase English letters.

**For example:**

| Test | Result |
|---|---|
| `print(checkPangram('thequickbrownfoxjumpsoverthelazydog'))` | `true` |
| `print(checkPangram('arvijayakumar'))` | `false` |

**Answer:** (penalty regime: 0 %)

Reset answer

```
def checkPangram(sentence):
    if len(set(sentence)) == 26:
        return "true"
    else:
        return "false"

```

| | Test | Expected | Got | |
|---|---|---|---|---|
| ✓ | `print(checkPangram('thequickbrownfoxjumpsoverthelazydog'))` | `true` | `true` | ✓ |
| ✓ | `print(checkPangram('arvijayakumar'))` | `false` | `false` | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

The program must accept **N** series of keystrokes as string values as the input. The character **^** represents undo action to clear the last entered keystroke. The program must print the string typed after applying the undo operations as the output. If there are no characters in the string then print **-1** as the output.

**Boundary Condition(s):**

1 <= N <= 100
1 <= Length of each string <= 100

**Input Format:**

The first line contains the integer N.
The next N lines contain a string on each line.

**Output Format:**

The first N lines contain the string after applying the undo operations.

**Example Input/Output 1:**

Input:

3
Hey ^ goooo^^glee^
lucke^y ^charr^ms
ora^^nge^^^^

Output:

Hey google
luckycharms
-1

**Answer:** (penalty regime: 0 %)

```
N = int(input())

for _ in range(N):
    s = input()
    result = []

    for ch in s:
        if ch == '^':
            if result:
                result.pop()
        else:
            result.append(ch)

    if result:
        print(''.join(result))
```

```
    else:
        print("-1")

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | 3<br>Hey ^ goooo^^glee^<br>lucke^y ^charr^ms<br>ora^^nge^^^^ | Hey google<br>luckycharms<br>-1 | Hey google<br>luckycharms<br>-1 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Find if a String2 is substring of String1. If it is, return the index of the first occurrence. else return -1.

**Sample Input 1**

thistest123string

123

**Sample Output 1**

8

**Answer:** (penalty regime: 0 %)

```
String1 = input()
String2 = input()

print(String1.find(String2))

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✓ | thistest123string<br>123 | 8 | 8 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.